
Inference Can’t Buy Knowledge: Agent Orchestration and Extended Reasoning Hit the Same Ceiling on Biology Research Tasks

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Abstract

1 Agentic systems for biology usually report gains over a single model call while
2 spending several calls. We ask where such gains come from on objectively scored
3 biology research tasks (LAB-Bench and LABBench2: protocol troubleshooting,
4 sequence manipulation, database and literature questions). At a matched budget
5 of three calls, we compare independent sampling with majority vote, sequential
6 self-critique, and a two-scientist debate with an adjudicator, across three open
7 and API models on 150 pre-registered questions scored without an LLM judge,
8 and we contrast these with simply letting frontier models reason longer. Self-
9 critique never beats majority voting (pooled -4.5 points) and breaks at least as
10 many answers as it repairs. Debate helps only where the model cannot reason on
11 its own: it adds $+8.7$ points for Gemma with thinking disabled, but -2.0 and -5.3
12 for models that reason, and no reliable gain for frontier models at high reasoning
13 effort. Raising reasoning effort rarely helps either: GPT-OSS generates $25\times$ more
14 tokens at high effort for no gain, and across six model families accuracy varies
15 little while spending varies by nearly two orders of magnitude. The dominant
16 factor is the question itself: across systems, question identity explains 56% of the
17 variance in correctness, while the choice of model, strategy and reasoning effort
18 explains 0.6%. On these tasks, additional inference, however organised, does not
19 substitute for missing knowledge.

20 1 Introduction

21 LLM agents now plan experiments, analyse data and generate hypotheses in chemistry and biology
22 [1, 2, 5, 6, 8, 15]. Their gains are typically reported against a single-call baseline, although the agent
23 makes many calls. Because repeated sampling with a vote already improves accuracy [3, 12, 18],
24 such comparisons conflate *what the interaction adds* with *how much inference was spent* [9].

25 For anyone building life-science agents, this is a design question as much as an evaluation one:
26 should the budget go into richer orchestration (critics, debating scientists) or somewhere else? We
27 run a small, controlled test of this confound on biology research questions with deterministic an-
28 swers. Every orchestrated strategy gets exactly the budget of the voting baseline (three calls, same
29 model, same prompts, same decoding). We then ask a second question the agent literature rarely
30 asks: does letting a strong model reason longer in a *single* call reach the same point?

31 **Findings.** (1) At matched budget, sequential critique never improves on majority voting, and
32 multi-agent debate improves on it only for a model whose reasoning is switched off. (2) Critique
33 never helps: it breaks at least as many correct answers as it repairs, and for models that reason it

Table 1: Accuracy (%) at a matched budget of three calls; DIRECT uses one call. Δ is DEBATE $-$ VOTE in points with a paired bootstrap 95% CI.

Model	n	DIRECT	VOTE	CRITIQUE	DEBATE	Δ [95% CI]
GPT-OSS-120B	150	46.7	49.3	42.7	47.3	-2.0 [-8.7, +4.7]
Gemma 4 31B	150	48.7	48.7	47.3	57.3	+8.7 [+2.0, +15.3]
Flash-Lite	75	61.3	62.7	56.0	57.3	-5.3 [-13.3, +1.3]

is most harmful on exact sequence manipulation. (3) Thinking longer is not a way out: higher reasoning effort helps only a model that otherwise does not reason at all, and then only up to the level its low-effort debate already reached; stacking orchestration on top adds nothing reliable. (4) Most strikingly, whether a system is right depends overwhelmingly on *which question* it faces and barely on *how* it spends inference: every system solves the same questions and fails the same questions (Fig. 2). The remaining errors look like missing knowledge, not missing deliberation.

2 Setup

Tasks. We use four LAB-Bench categories [10] (ProtocolQA, SeqQA, DbQA, LitQA2; multiple choice with shuffled options and no abstain option) and the deterministic subset of LABBench2 SeqQA2 [11] (open-ended sequence computations with unique answers, scored by the official validators, with input files injected into the prompt). We exclude figure/table tasks (text-only models) and prompts over 6,000 characters. With a fixed seed we draw 30 questions per category (150 total), plus 10 pilot questions that are excluded from all results. No LLM judge is used anywhere.

Strategies (Fig. 3 in App. A). DIRECT makes one call. VOTE makes three independent calls and takes the majority answer, breaking ties by self-reported confidence. CRITIQUE solves, critiques the solution, then revises. DEBATE has a molecular biologist and a skeptical reviewer solve independently, and a third call adjudicates between them. VOTE, CRITIQUE and DEBATE each use exactly three calls. Only visible outputs are passed between calls.

Models. The main study uses GPT-OSS-120B [13] (`reasoning_effort=low`), Gemma 4 31B [16] (thinking off, the only non-default setting the API accepts), and Gemini 3.5 Flash-Lite (thinking low), all through free API tiers. Decoding uses temperature 0.6, fixed per-call seeds and a 2,048-token cap. GPT-OSS and Gemma answer all 150 questions and Flash-Lite a fixed 75-question subset. For the reasoning-effort analysis we additionally run GPT-6 Luna, GPT-6 Sol and GPT-5.6 Terra (all strategies, high effort; Luna also at low effort), GPT-OSS at high effort (all strategies, served by Ollama; a low-effort DIRECT run on the same host matches the main run’s accuracy) and Flash-Lite (DIRECT at high thinking), all on the first 30 questions in the fixed run order (6 per category).

Pre-registration. Task IDs, prompts, decoding, hypotheses and the analysis script were committed before the main run. The primary contrast is DEBATE $-$ VOTE, tested with a paired bootstrap over questions (10,000 resamples) and an exact McNemar test. Post-freeze deviations are listed in App. B: rate-limit workarounds, a second inference provider for 22 GPT-OSS questions (each question served entirely by one provider), and the exploratory reasoning-effort arm.

3 Results

Orchestration beats voting only when the model cannot think. DEBATE improves on VOTE for Gemma, whose thinking is disabled (+8.7 points, 95% CI [+2.0, +15.3], McNemar $p = 0.015$), and the gain survives excluding every question with a truncated call. For the two models that reason, the interval includes zero ($p = 0.7$ and 0.29 ; for GPT-OSS the estimate is unchanged at -1.6 when the 22 questions served by the second provider are excluded). Pooled over all 375 model-question pairs, DEBATE $-$ VOTE is $+1.6$ [$-2.4, +5.6$] points and CRITIQUE $-$ VOTE is -4.5 [$-9.3, +0.3$]. CRITIQUE is never better than VOTE ($-6.7, -1.3, -6.7$ points). Voting itself barely improves on a single call ($+2.7, +0.0, +1.3$ points), even though the three samples disagree on 38%, 42% and 27% of questions.

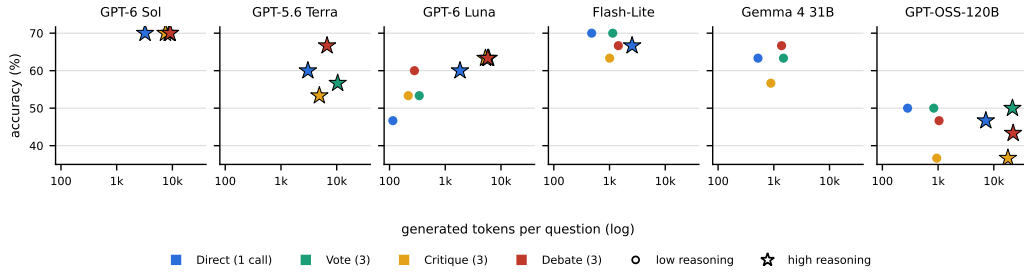


Figure 1: Accuracy vs. generated tokens per question on the same 30 questions. Circles are low reasoning and stars are high reasoning; colour marks the strategy. Across nearly two orders of magnitude in generated tokens, all ways of spending inference land on a narrow per-model plateau.

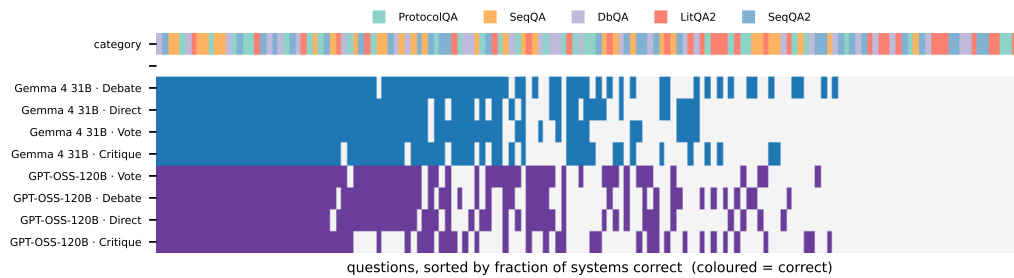


Figure 2: Correctness of every model $\times$ strategy system (rows) on every question answered by all of them (columns, sorted by the fraction of systems correct). The staircase shows that systems solve and fail the *same* questions and differ mainly in a shared boundary. Question identity explains 56% [49, 62] of the variance in correctness; the system explains 0.6% [0.3, 1.9].

Critique breaks at least as often as it repairs. Relative to the vote, CRITIQUE rescues 24%, 13% and 25% of wrong answers but corrupts 38%, 16% and 26% of correct ones. For the two models that reason, the largest losses are on exact sequence manipulation (SeqQA: -23 and -20 points vs. VOTE for GPT-OSS and Flash-Lite; +10 for Gemma), where a fluent but wrong critique talks the model out of a correct computation.

Thinking longer mostly does not help either. On the same 30 questions (Fig. 1), raising GPT-OSS to high reasoning effort multiplies generated tokens by $25\times$ (284 to 7,191 per question) while accuracy stays flat (50% to 47%). For Flash-Lite the result is the same (70% to 67%, $5\times$ tokens). The one model that gains is GPT-6 Luna, which at low effort uses no reasoning tokens (47% to 60%). Its high-effort single call lands exactly where its low-effort DEBATE already was (60%), consistent with debate acting as a substitute for reasoning. Orchestration on top of high effort adds nothing reliable. GPT-6 Sol scores 70% with all four strategies, and GPT-5.6 Terra ranges from 53% (CRITIQUE) to 67% (DEBATE) with no paired difference significant. Within each model, accuracy moves by a few questions while generated tokens vary by up to two orders of magnitude.

The question, not the method, determines the outcome. A two-way decomposition of the 0/1 correctness matrix (systems $\times$ questions) attributes 56% of the variance to the question and 0.6% to the system for the main models (Fig. 2). Across all 33 systems on the shared 30 questions, including frontier models at high reasoning, the split is 47% vs. 3.9%. Of the 32 questions that no main system solves, 18 require literature or database recall (LitQA2, DbQA) and 9 are open-ended sequence computations (SeqQA2), failures that point to missing retrieval and code execution rather than missing deliberation.

97 4 Implications for building life-science agents

98 **Benchmark against a compute-matched vote.** A multi-call agent should beat independent sam-
99 pling with the same number of calls, not a single call [9]. Most of our orchestration gains vanish
100 under that baseline. **Spend the harness budget on tools, not talk.** The questions every system
101 fails need literature or database recall, or exact sequence computation. Those are the failures that
102 retrieval, database APIs and code execution address, and that extra conversations between copies of
103 the same model cannot. **A stronger generalist does not remove the ceiling.** Frontier models at high
104 reasoning effort plateau at the same per-question boundary. This suggests that on knowledge-bound
105 biology tasks, access to specialised tools and data matters more than a stronger or longer-thinking
106 backbone. Our results agree with compute-matched analyses of debate on general reasoning bench-
107 marks [4, 14, 17, 19] and with evidence that intrinsic self-correction is unreliable [7]. We extend
108 them to biology, to sequential critique, and to the trade-off between orchestration and reasoning
109 effort.

110 **Limitations.** Models had no tools or retrieval, so DbQA and LitQA2 measure recall; orchestration
111 with tools may behave differently. The budget is fixed at three calls. Frontier models were run
112 through the Codex CLI, which adds its own system prompt and does not expose temperature or
113 seeds; they cover 30 questions, so their comparisons are exploratory. Gemma could only run with
114 thinking off. One prompt set per strategy was used, and 150 questions give limited power for effects
115 under ~ 10 points.

116 5 Conclusion

117 On objectively scored biology research tasks, how a model spends extra inference (voting, self-
118 critique, multi-agent debate or longer reasoning) barely changes which questions it gets right: ques-
119 tion identity explains 56% of the variance in correctness and the choice of system 0.6%. Debate
120 helps only a model that cannot otherwise reason, and critique never helps. For builders of life-
121 science agents, the lever is knowledge access through retrieval, databases and computational tools,
122 not more elaborate conversations between copies of the same model.

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A Prompts

All prompts are single user messages. {QUESTION} is the question (with protocol, options or injected files) and {ANSWER INSTRUCTION} is the format line for multiple choice or the official SeqQA2 suffix.

Direct / Independent×3 / Critique call 1 (solve)

Solve the following biology research question.

{QUESTION}

{ANSWER INSTRUCTION}

Respond in exactly this format and keep it short:

<answer>YOUR ANSWER</answer>

CONFIDENCE: <probability between 0 and 1 that your answer is correct>

JUSTIFICATION: <at most two sentences>

201 **Critique call 2 (critique)**

202 Below are a biology research question and a proposed solution. Act as a critical reviewer: identify any
203 scientific, computational or reasoning error in the solution. If you think the answer is wrong, say which
204 answer you believe is correct and why. If you find no error, say so.

205
206 {QUESTION}

207
208 Proposed solution:
209 {CALL 1 OUTPUT}

210
211 Respond with at most four sentences, starting with "CRITIQUE:".

212 **Critique call 3 (revise)**

213 Below are a biology research question, a first attempt at solving it, and a reviewer's critique of that
214 attempt. Either may contain mistakes. Considering both, produce the final answer.

215
216 {QUESTION}

217
218 First attempt:
219 {CALL 1 OUTPUT}

220
221 Reviewer's critique:
222 {CALL 2 OUTPUT}

223
224 {ANSWER INSTRUCTION}

225
226 Respond in exactly this format and keep it short:
227 <answer>YOUR ANSWER</answer>
228 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
229 JUSTIFICATION: <at most two sentences>

230 **Debate call 1 (researcher)**

231 You are a molecular biology researcher. Solve the following biology research question independently.

232
233 {QUESTION}

234
235 {ANSWER INSTRUCTION}

236
237 Respond in exactly this format and keep it short:
238 <answer>YOUR ANSWER</answer>
239 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
240 JUSTIFICATION: <at most two sentences>

241 **Debate call 2 (skeptic)**

242 You are a skeptical scientific reviewer. Solve the following biology research question independently, looking
243 especially for subtle experimental, biological or computational errors and traps.

244
245 {QUESTION}

246
247 {ANSWER INSTRUCTION}

248
249 Respond in exactly this format and keep it short:
250 <answer>YOUR ANSWER</answer>
251 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
252 JUSTIFICATION: <at most two sentences>

253 **Debate call 3 (adjudicator)**

254 Two scientists independently analysed the biology research question below. Adjudicate between their analyses
255 (either or both may be wrong) and submit the final answer.

256
257 {QUESTION}

258

259 Analysis 1 (molecular biology researcher):
 260 {CALL 1 OUTPUT}
 261
 262 Analysis 2 (skeptical scientific reviewer):
 263 {CALL 2 OUTPUT}
 264
 265 {ANSWER INSTRUCTION}
 266
 267 Respond in exactly this format and keep it short:
 268 <answer>YOUR ANSWER</answer>
 269 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
 270 JUSTIFICATION: <at most two sentences>

271 B Pre-registration deviations

272 All task IDs, prompts, decoding settings and the primary analysis were frozen after a 10-question
 273 pilot and before any main-set call. Changes made after the freeze: (i) throttle and retry fixes for an
 274 undocumented 150 requests/hour provider limit (no completed call was re-run); (ii) 22 GPT-OSS
 275 questions served in full by a second provider of the same model (identical prompts, seeds and cap;
 276 excluding them leaves DEBATE – VOTE at -1.6), with partially served questions discarded and
 277 re-run on the primary provider; (iii) the exploratory reasoning-effort arm on the first 30 questions,
 278 including frontier models served through the Codex CLI with web search, code execution and sub-
 279 agents disabled (a first attempt with web search enabled was discarded unanalysed), and GPT-OSS
 280 at high effort on a second host.

281 C Additional results

Category	Gemma					Lite					OSS				
	D	I	C	M	Disagree	D	I	C	M	Disagree	D	I	C	M	Disagree
DbQA	56.7	53.3	40.0	56.7	33.3	40.0	40.0	40.0	33.3	26.7	46.7	46.7	43.3	36.7	36.7
LitQA2	33.3	30.0	33.3	36.7	46.7	60.0	53.3	53.3	46.7	26.7	20.0	30.0	40.0	36.7	43.3
ProtocolQA	56.7	56.7	60.0	70.0	20.0	66.7	80.0	86.7	73.3	26.7	56.7	50.0	43.3	46.7	33.3
SeqQA	56.7	56.7	66.7	73.3	46.7	73.3	73.3	53.3	66.7	13.3	63.3	66.7	43.3	60.0	33.3
SeqQA2	40.0	46.7	36.7	50.0	63.3	66.7	66.7	46.7	66.7	40.0	46.7	53.3	43.3	56.7	43.3

Table 2: Per-category accuracy (D=Direct, I=Independent×3, C=Critique, M=Multi). Numbers are percentages.